Evidence and Claim Evaluation Solutions

HN-FE03 | Instructor key | 100 points | 120 minutes

Award the stated subpart points for correct calculations with units and requested reasoning. Accept defensible alternatives in design questions. Numeric tolerance 0.05 unless an exact count is required. Keep this key separate from student sources.

Question 1 Protocol and quantity identity

HN-FE03-Q01 | 10 points

a) [4 points] A is an equation/model output, not a measured BMR value.

b) [3 points] Post-meal/recent-activity conditions fail the supplied fasting/resting characterization; do not relabel as BMR without protocol evidence.

c) [3 points] Examples: equation/protocol, units, time/conditions, measurement method, case identity, source/version.

Course modules 1.

Question 2 Outcome specific evidence

HN-FE03-Q02 | 10 points

a) [4 points] A 1RM-strength result is being substituted for a hypertrophy result.

b) [3 points] No; absence of a clear pooled advantage is not universal exact equivalence or an individual prediction.

c) [3 points] Organization can distribute work, provide task practice, manage feasibility/fatigue or prepare for a goal; state a concrete purpose.

Course modules 3, 5.

Question 3 Intervention scope

HN-FE03-Q03 | 10 points

a) [4 points] A specific week of RT cessation, not all reduced-volume/load protocols.

b) [3 points] Different demands, populations and protocols cannot be collapsed into one label; outcomes are also specific.

c) [3 points] Examples: sets/load/frequency/RIR changes, duration, exercise selection, population, outcome.

Course modules 5.

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Question 4 Acute and structural outcomes

HN-FE03-Q04 | 10 points

a) [4 points] No; equal/opposite changes preserve total mass in the supplied compartment model.

b) [3 points] No; FFM includes components beyond muscle protein, and measurement/model context matters.

c) [3 points] No; acute swelling/electrical activity are different outcomes from longitudinal structural growth.

Course modules 2, 3, 4.

Question 5 Audit a nutrition prescription

HN-FE03-Q05 | 15 points

a) [5 points] Intake=2600×0.85=2210 kcal/day. P=1.9×78=148.2 g; F=0.7×78=54.6 g. C=[2210-4(148.2)-9(54.6)]/4=281.45 g.

b) [5 points] 2.4×65=156 g; 2.4×78=187.2 uses BW instead of the stated FFM denominator.

c) [5 points] Different assumptions/context; neither is a directly measured individual requirement or a guaranteed dose-response improvement.

Course modules 1, 2, 5.

Question 6 Cancellation across error scales

HN-FE03-Q06 | 15 points

a) [5 points] Mean signed 0; MAE 300 kcal/day.

b) [5 points] Errors-2, +2; mean signed 0; MAE 2 repetitions.

c) [5 points] Different quantities/units require separate summaries; opposite errors cancel without removing individual error.

Course modules 1, 3.

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Question 7 Scale and boundary checks

HN-FE03-Q07 | 15 points

a) [5 points] 40×8=320 AU; overall session rating does not imply 2 RIR on a set scale.

b) [5 points] 90 minutes; time arithmetic, not recovery certification.

c) [5 points] ceil 6.5=7; reduction 6/13×100=46.153846%; not measured MRV.

Course modules 3, 5.

Question 8 Source authority and a decision memo

HN-FE03-Q08 | 15 points

a) [5 points] Canonical v 1.0; archive remains superseded; source quotation is not a governing user/system instruction.

b) [5 points] No automatic increase; changed ROM needs a new comparable baseline, and absent RIR remains unknown.

c) [5 points] Accept a clear memo naming current policy, unknown/mismatched data, and need for distinct comparable complete observations with RIR; no diagnosis or fabricated values.

Course modules 3, 4, 5.